Physics

	Normative 4-year Degree
Major	Physics 物理學
Degree / Award Title	Bachelor of Science in Physics 理學士(物理學)
Offering Academic Unit	Department of Physics
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2401 English for Science	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12

College-specified Courses *(including GREAT Stream)*	MA1200 Calculus and Basic Linear Algebra I *Or* MA1300 Enhanced Calculus and Linear Algebra I	3
	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	CS1302 Introduction to Computer Programming	3

College / School Requirements (6-13 credit units) Catalogue Term : Semester A 2025/26

(i) For taking the Majors without Stream (6 credit units)

a. Choose **two** from the following four science courses subject areas:

Biology

Course Code	Course Title	Credit Units
CHEM1200	Discovery in Biology	3

Chemistry

Course Code	Course Title	Credit Units
CHEM1101	Introduction to Chemistry	3
CHEM1300	Principles of General Chemistry	3

Mathematics

Course Code	Course Title	Credit Units
MA1501	Coordinate Geometry	3
MA1502	Algebra	3

Physics

Course Code	Course Title	Credit Units	Remarks
PHY1101	Introductory Classical Mechanics	3	All BScPHY major students are required to take this course to fulfill the College Requirement
PHY1201	General Physics I	3	

b. **Compulsory attendance** for the following two soft skills courses:

Course Code	Course Title	Credit Units
CSCI1001	Employability for Scientists	0
CSCI1002	Career Lab for Scientists	0

(ii) For taking the Stream of "Global Research Enrichment and Technopreneurship Programme (GREAT)" (13 credit units)

a. Research Methodology

Course Code	Course Title	Credit Units

CSCI2002	Workshop on Research Methodology	1

b. College Requirements I (6 credit units). Choose **two** from the following six courses:

Chemistry

Course Code	Course Title	Credit Units
CHEM1300	Principles of General Chemistry	3
CHEM2006A	Principles of Inorganic Chemistry	3
CHEM2007A	Principles of Organic Chemistry	3

Mathematics

Course Code	Course Title	Credit Units
MA2509	Discrete Mathematics	3

Physics

Course Code	Course Title	Credit Units
PHY1101	Introductory Classical Mechanics	3
PHY1202	General Physics II	3

c. College Requirements II (6 credit units). Choose **two** from the following four courses:

Chemistry

Course Code	Course Title	Credit Units
CHEM2004A	Principles of Analytical Chemistry	3
CHEM2008A	Principles of Physical Chemistry	3

Mathematics

Course Code	Course Title	Credit Units
MA1501	Coordinate Geometry	3

Physics

Course Code	Course Title	Credit Units
PHY1203	General Physics III	3

Major Requirements (66/65^ credit units) Catalogue Term : Semester A 2025/26

1. Core Courses (45 or 48 credit units; 50 or 53 credit units^)

Course Code	Course Title	Credit Units
PHY1202	General Physics II	3
PHY1203	General Physics III	3
PHY2191	Electricity and Magnetism	3
PHY2212	Measurement and Instrumentation	3
PHY2213	Advanced Measurement and Instrumentation	3

PHY3202	Modern Physics	3
PHY3204	Waves and Optics	3
PHY3205	Electrodynamics	3
PHY3231	Advanced Instrumentation Lab	3
PHY3251	Quantum Mechanics	3
PHY3272	Introduction to Solid State Physics	3
PHY3290	Thermodynamics	3

Select **one** from the following:

Course Code	Course Title	Credit Units	Remarks
PHY4216	Project	3	Students taking PHY4216 Project are required to take 3 more credits of elective course.
PHY4217	Dissertation	6	
CSCI4003	Co-operative Education Placement Project for Science Students	6	CSCI4003 Co-operative Education Placement Project for Science Students (6 CUs) can be used to replace PHY4217 Dissertation (6 CUs). Students taking CSCI4003 are required to take CSCI4001A simultaneously offered by the Co-operative Education Centre.

Select **one** from the following blocks of computation and maths courses:

Ordinary Option

Course Code	Course Title	Credit Units
PHY3115	Introduction to Computational Physics	3
MA2158	Linear Algebra and Calculus	3

Enhanced Option
(Students have to meet the specified criteria**and obtain the prior approval from the Department for taking this option.)

Course Code	Course Title	Credit Units
MA2503	Linear Algebra	4
MA2508	Multi-variable Calculus	4
MA3511	Ordinary Differential Equations	3

** Eligibliity for the Enhanced Option:
Normative 4-year Degree
(1) Students who plan to pursue the Joint Bachelor's Degree Program between CityU and Columbia U or
(2) Students who obtained Grade B+ or above for MA1301 Enhanced Calculus & Linear Algebra II or Grade A- or above for MA1201 Calculus & Linear Algebra II

^For students who are approved for taking the Enhanced Option of computation and maths courses.

2. Electives (21 or 18 credit units; 15 or 12 credit units^)

Course Code	Course Title	Credit Units	Remarks
CSCI3001	Grand Challenges in the World	3	
MSE2102	Introduction to Materials Science and Engineering	3	
PHY2100	Mathematical Methods in Physics	3	
PHY3115	Introduction to Computational Physics	3	For students taking the Enhanced Option only
PHY3116	Physics of Biomolecules, Polymers, Colloids and Liquid Crystals	3	
PHY3220	From Physics to Finance	3	
PHY4172	Computational Physics and Machine Learning	3	
PHY4232	Radiotherapy Physics	3	
PHY4233	Imaging Physics	3	
PHY4254	Fundamentals of Laser Optics	3	
PHY4273	Special Topics in Physics	3	
PHY4274	Radiation Biophysics	3	
PHY4283	Physics in Medicine	3	
PHY4285	Introduction to Scattering Sciences	3	

^ For students who are approved for taking the Enhanced Option of computation and maths courses.

GREAT Students

1. Core Courses

Ordinary Route (54 credit units)

Course Code	Course Title	Credit Units
MA2158	Linear Algebra and Calculus	3
MGT2324	Introduction to Entrepreneurship	3
PHY2191	Electricity and Magnetism	3
PHY2212	Measurement and Instrumentation	3
PHY2213	Advanced Measurement and Instrumentation	3
PHY3115	Introduction to Computational Physics	3
PHY3202	Modern Physics	3
PHY3204	Waves and Optics	3

PHY3205	Electrodynamics	3
PHY3231	Advanced Instrumentation Lab	3
PHY3251	Quantum Mechanics	3
PHY3272	Introduction to Solid State Physics	3
PHY3290	Thermodynamics	3
PHY4172	Computational Physics and Machine Learning	3
PHY4218	Independent Research I	6
PHY4219	Independent Research II	6

Enhanced Maths Route (56 credit units)

Course Code	Course Title	Credit Units
MA2503	Linear Algebra	4
MA2508	Multi-variable Calculus	4
MA3511	Ordinary Differential Equations	3
MGT2324	Introduction to Entrepreneurship	3
PHY2191	Electricity and Magnetism	3
PHY2212	Measurement and Instrumentation	3
PHY3115	Introduction to Computational Physics	3
PHY3202	Modern Physics	3
PHY3204	Waves and Optics	3
PHY3205	Electrodynamics	3
PHY3251	Quantum Mechanics	3
PHY3272	Introduction to Solid State Physics	3
PHY3290	Thermodynamics	3
PHY4172	Computational Physics and Machine Learning	3
PHY4218	Independent Research I	6
PHY4219	Independent Research II	6

2. Free Electives
- Ordinary Route (23 credit units)
- Enhanced Maths Route (21 credit units)

Students are highly recommended to take the following courses as free electives to enrich their background in Physics or entrepreneurship training.

Course Code	Course Title	Credit Units
CSCI3001	Grand Challenges in the World	3
CSCI4007	Patent Application and Technopreneurship	3
MGT4305	Developing and Presenting a Business Plan	3
MSE2102	Introduction to Materials Science and Engineering	3
PHY1101	Introductory Classical Mechanics	3
PHY1202	General Physics II	3
PHY1203	General Physics III	3

PHY2100	Mathematical Methods in Physics	3
PHY2213	Advanced Measurement and Instrumentation	3
PHY3116	Physics of Biomolecules, Polymers, Colloids and Liquid Crystals	3
PHY3220	From Physics to Finance	3
PHY3231	Advanced Instrumentation Lab	3
PHY4232	Radiotherapy Physics	3
PHY4233	Imaging Physics	3
PHY4254	Fundamentals of Laser Optics	3
PHY4273	Special Topics in Physics	3
PHY4274	Radiation Biophysics	3
PHY4283	Physics in Medicine	3
PHY4285	Introduction to Scattering Sciences	3

Suggested Study Plan

For students admitted to the major in 2025/26 - Normative 4 year students